

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 03-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-29** **Max. Marks: 180**

03-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2022-P2)_GTA-29_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCE-2022-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

1. For $x \in \mathbb{R}$, then the number of real solutions of the equation $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is _____.

2. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c,\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{2b}{a}$ is _____.

3. Let the function $f:(0,\pi) \to \mathbb{R}$ be defined by $f(\theta) = (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^4$

Suppose the function f has a local minimum at θ precisely when $\theta \in \{\lambda_1\pi, ..., \lambda_r\pi\}$,

Where $0 < \lambda_1 < ... < \lambda_r < 1$. Then twice of reciprocal of $\lambda_1 + ... + \lambda_r$ is ____.

4. Let the mean and variance of four numbers $3, 7, x$ and $y (x > y)$ be 5 and 10 respectively.

Then half of mean of four numbers $3 + 2x, 7 + 2y, x + y$ and $x - y$ is___.

5. If $(1 + 3 + 5 + ... + a) + (1 + 3 + 5 + ...b) = (1 + 3 + 5.... + c)$, such that (i) $a + b + c = 57$ and

(ii) $a < 10$, then the value of $(2b - 2c + a)$ is equal to

6. $\frac{1}{2026\pi}\int_{-2026}^{2026} \frac{dx}{\sqrt{\{x + 2027\}\{-x + 2028\}}} =$ __ (where $\{.\}$ fractional part function)

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7. If $\cos^{-1}x \in \left(0, \frac{\pi}{2}\right)$, then the value of

$$\left[\tan^2\left(\frac{\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)+\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)}{2}\right)\right]$$ is equal to_____

[.] denotes greatest integer function)

8. If $\int \frac{\cos 8x - \cos 7x}{1+2\cos 5x}dx = p\sin 3x + q\sin 2x + C, p,q \in R$, then the value of

$\left(\left|[2p]\right|+\left|[3q]\right|\right)+\left|[-4p]+[5q]\right|$ is ___,where [.] represents greatest integer function.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

9. For integers m,n with $-1 \le m,n \le 1$, consider $A_{m,n} = \begin{pmatrix} m & 1 & 1 \\ 1 & n & 1 \\ 1 & 1 & m \end{pmatrix}$.

We call (m,n) good if $A_{m,n}$ is invertible and $A_{m,n}^{-1}$ has all integer entries. Which of the following statement(s) is/are correct?

A) Exactly two ordered pairs (m,n) in $[-1,1]\times[-1,1]$ are good.

B) $(m,n)=(2,1)$ is good

C) There exists a good pair with $\det(A_{m,n})=1$.

D) The sum of elements of $A_{m,n}$ over all good pair(s) equals 1

10. If $\hat{a}, \hat{b}, \hat{c}$ are unit vectors such that $\hat{a}+\hat{b}+\hat{c}=\bar{O}$ and x,y,z be distinct integers, then $\left|x\hat{a}+y\hat{b}+z\hat{c}\right|$ cannot be equal to

A) 1 **B)** $\sqrt{2}$ **C)** 2 **D)** 3

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11. If $f(x)=\begin{cases} x^m \cos\left(x-\frac{1}{x}\right), & x>0 \\ 0, & x\le 0 \end{cases}$, then which of the following is/are correct?

A) $\forall m\in(-\infty,0]$, $f(x)$ is discontinuous at $x=0$

B) $\forall m\in(0,\infty)$, $f(x)$ is differentiable at $x=0$

C) $\forall m\in(1,\infty)$, $f(x)$ is differentiable at $x=0$

D) $\forall\ m\in(0,1]$, $f(x)$ is continuous but not differentiable at $x=0$

12. If $lx+my+1=0$ is common tangent of curves $x^2=4y$, $x^2+\frac{y^2}{3}=\frac{1}{4}$ and $y^2=4ax, a>0$, then which of the following is/are correct?

A) $\left(4l^2-a\right)$ is odd prime number

B) $\left(3m^2+3a\right)$ is even composite number

C) $\left(\frac{5l-a}{4m}\right)$ is neither prime nor composite number

D) $\left(2a^2+3l^2+4m\right)$ is odd composite number

13. The number of selections of fruits from 4 mangoes, 3 oranges, 2 papayas and 2 pineapples will be [fruits of same type are treated as identical]

A) 179 (when atleast one fruit is chosen)

B) 175 (atleast 2 fruits are chosen)

C) 15 (at most 2 fruits are chosen with atleast one fruit)

D) 15 (none or more fruits but not more than 2 fruits are chosen)

14. From point $A(-1,1,-1)$, the perpendicular is drawn to a plane P_1 whose foot is $(1,-1,1)$. From point $B(3,-1,-1)$, the perpendicular is drawn to a plane P_2 whose foot is $(1,2,1)$. P_3 is a plane containing the line of intersection of P_1 *and* P_2 and passing through the mid point of the line joining A and B. Then which of the following statement(s) is/are correct?

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A) Direction ratios of line of intersection of P_1 *and* P_2 can be $5,4,-1$

B) Equation of plane P_3 is $16x-19y+4z=12$

C) Equation of line of intersection of P_1 *and* P_2 is $\frac{x-15}{5}=\frac{y-12}{4}=\frac{z-0}{-1}$

D) The length of perpendicular from $(0,0,0)$ to plane P_3 is $\frac{2}{\sqrt{18}}$

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

15. If $A=\begin{bmatrix} 1+x^2-y^2-z^2 & 2(xy+z) & 2(zx-y) \\ 2(xy-z) & 1+y^2-z^2-x^2 & 2(yz+x) \\ 2(zx+y) & 2(yz-x) & 1+z^2-x^2-y^2 \end{bmatrix}$, then $|A|$ is equal to

A) $(1+xy+yx+zx)^3$ **B)** $\left(1+x^2+y^2+z^2\right)^3$

C) $(xy+yx+zx)^3$ **D)** $\left(1+x^3+y^3+z^3\right)^2$

16. A fair coin is tossed until one of the two sides occurs twice in a row, The Probability that the number of tosses required is even is

A) 1/3 **B)** 2/3 **C)** 1/4 **D)** 3/4

17. Value of $\sum_{k=1}^{100}\left(i^{k!}+\omega^{k!}\right)$, where $i=\sqrt{-1}$ and ω is non-real cube root of unity, is

A) $190+\omega$ **B)** $192+\omega^2$ **C)** $190+i$ **D)** $192+i$

18. Let $P(0,3),Q(0,5),R(x,0)$, then the value of $|x|$ for which $\angle PRQ$ is as large as possible:

A) $\sqrt{15}$ **B)** $\sqrt{20}$ **C)** $\sqrt{35}$ **D)** $\sqrt{40}$

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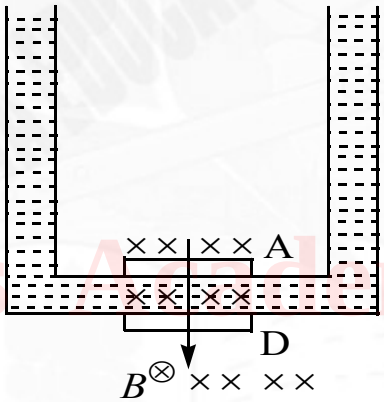
PHYSICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

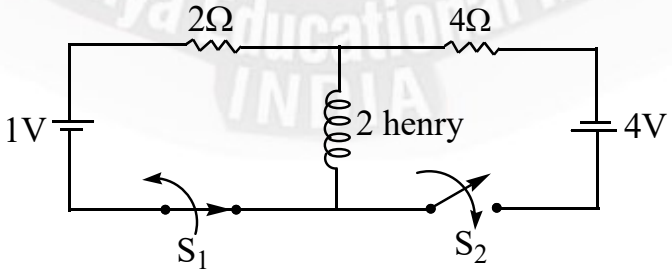
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Negative Marks: -1 In all other cases

19. Mercury is contained in U-tube of uniform square cross-section of side $a = 4\,mm$. Electrodes A and D are sealed inside as shown in the figure. An electric current $I = 10A$ passes between the electrodes from A and D and magnetic field $B = 5.44 Wb/m^2$ is applied across the horizontal arm perpendicular into the plane of the figure. Here density of mercury $= 13.6 \times 10^3\, kg/m^3$, $g = 10\, m/s^2$. Neglect surface tension. The difference (in mm) in mercury level in limbs of U tube is $\frac{400}{x}$. find x.



20. In the circuit shown in the figure, S_1 remains closed for a long time and S_2 remains open. Now S_2 is closed and S_1 is opened. Find out the $\left|\frac{di}{dt}\right|$ (in amp/sec) just after that moment, where i is the current through the inductance.

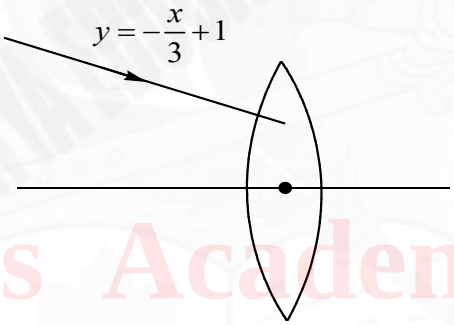


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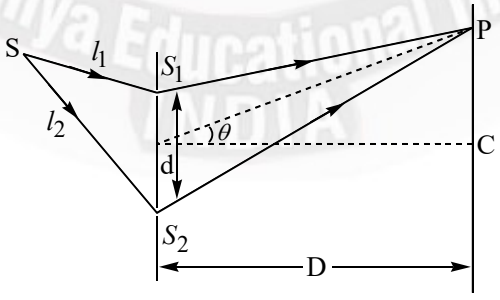
21. In a series LCR circuit, half power frequencies (frequency at which power dissipated is half the maximum power dissipated in the circuit) are found to be 90 rad/s and 40 rad/s. Find the value of $\frac{\omega_0}{10}$, where ω_0 is the resonance frequency.

22. When the voltage applied to an X-ray tube is increased from $V_1 = 10kV$ to $V_2 = 20kV$, the wavelength interval of K_α line and cut-off wavelength of continuous X-ray spectrum increases by a factor η. If atomic number of the element of which the tube's anode is made, is 29 then find the value of η. [Assume validity of Bohr's model for K_α x-rays]

23. A thin equi-convex lens of focal length 15 cm is placed at the origin with its optical axis coinciding with the $x-$axis. A ray light given by $y = -\frac{x}{3} + 1$ is incident on the lens. The equation of refracted ray is $y = -0.1mx + c$. (x and y are in cm). Find the value of m. [Assume equations of paraxial approximation]

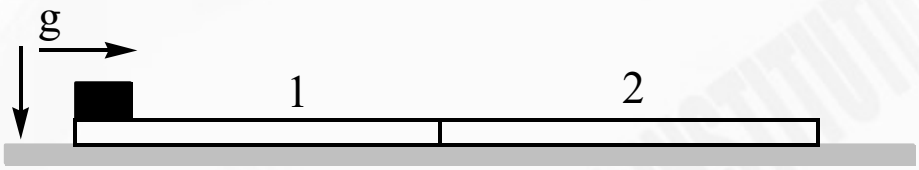


24. In a Young's doubt slit experiment the light source is at distance $l_1 = 20\mu m$ and $l_2 = 40\mu m$ from the slits. The light of wavelength $\lambda = 500nm$ incident on slits separated at a distance $d = 10\mu m$. A screen is placed at a distance D = 2m away from the slits as shown in the figure. If 10k maxima appear on the screen, then find the value of k. Round off your answer to the nearest integer, if required.

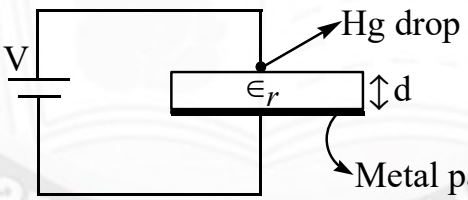


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25. On perfectly slippery ice, two identical boards lie in contact (figure). At the left edge a rough block is placed on the first board. When it was pushed, it reached the right edge of the second board and stayed on it. The final speed of second is n times the final speed of first board. What is the value of n? Weight and size of block is much less than mass and the length of the boards. [Round off n to nearest integer]



26. Bottom face of a horizontal dielectric slab is coated with a conducting paint and at centre of the top face is placed a small mercury drop, which does not wet the plate. Thickness of the slab is d and relative permittivity is ε_r. Now the switch is closed and the battery voltage is increased so gradually that the mercury drop spreads on the dielectric slab at negligible rate. At what battery voltage will the mercury drop cover the whole of the upper face of the dielectric? Surface tension of mercury is σ and initial surface area of the drop is negligible as compared to surface area of mercury when spread. If your answer is $V=\sqrt{\frac{x\sigma d}{\varepsilon_r\varepsilon_0}}$ write value of x. [Assume same σ for all interfaces]



SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
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27. A man of mass m on an initially stationary boat gets off of the boat by leaping to the left in an exactly horizontal direction. Immediately after the leap, the boat of mass M is observed to be moving to the right at speed v:

A) Work done by man on boat is $\frac{1}{2}(m)v^2$.

B) Increase in the mechanical energy of system of man and boat is $\frac{1}{2}\left(\frac{M^2}{m}+M\right)v^2$.

C) Velocity of centre of mass of system is v.

D) Increase in kinetic energy of man is $\frac{1}{2}\frac{M^2}{m}v^2$.

28. A particle of mass 1 kg moves along $x-$ axis in a conservative force field. It's potential energy is given by $U(x)=2x^3-9x^2+12x$, where all quantities are in S.I.unit. Then select correct statement.

A) Equilibrium positions are $x=1m$ and $x=2m$.

B) At $x=0$ force is along $(-x)$ direction.

C) Particle can perform oscillatory motion about $x=2m$.

D) For performing oscillatory motion, total mechanical energy should be less the $4J$.

29. Supercooling is state where liquids do not solidify even below their normal freezing point. Supercooled water can remain in the liquid state as low as -40^oC. If a small piece of ice is dropped then water suddenly freezes. If super cooled water at -10^oC suddenly starts freezing then in a very short interval a mixture of water and ice both at 0^oC is formed. If 1 kg of water freezes then 334 kJ of energy is released. Specific heat of ice $=2.1kJ\,kg^{-1}K^{-1}$, specific heat of water (in supercooled state also) $=4.2\,kJ\,kg^{-1}K^{-1}$, Latent heat of ice at $0^oC=334\,kJ\,kg^{-1}$. Consider 1 kg of water and mark the correct statements:

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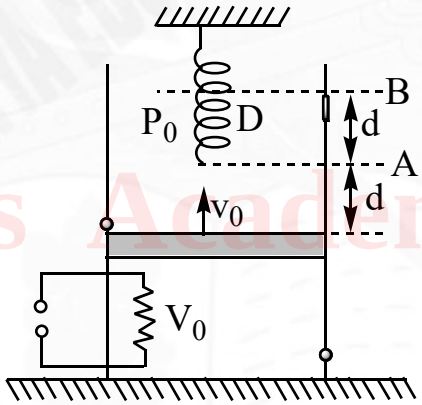
A) Latent heat of fusion at $-10^0 C$ is nearly $313 KJKg^{-1}$

B) Latent heat of fusion at $-10^0 C$ *is* $334 kJ\ kg^{-1}$.

C) If suspercooled water suddenly freezes at $-10^0 C$ then approximately 0.67Kg of ice is formed.

D) If suspercooled water suddenly freezes at $-10^0 C$ then approximately 0.13 Kg of ice is formed.

30. A mass less piston, which can move without friction, closes a sample of Helium in a vertical, thermally insulated cylinder, which is closed at its bottom, and the cross-section of which is $A = 2dm^2$.Above the piston there is a fixed stand to which an unstretched spring of spring constant 2000N/m is attached , whose bottom end is at a distance of 2dm from the piston when the volume of the gas is $V_0 = 8dm^3$. The external pressure $P_0 = 10^5 Pa$ & $g = 10m/s^2$.The gas confined in the cylinder is heated with an electric heating element .Mark the correct statement (s): $\left(1dm = 10^{-1} m\right)$



A) Internal energy change of helium till the piston reaches the spring is 1000J

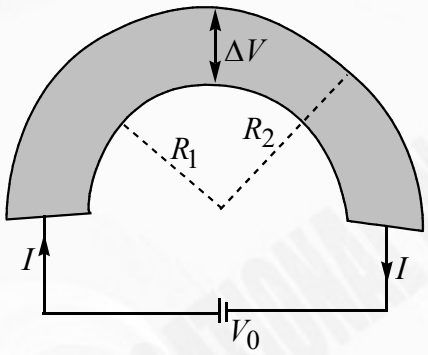
B) Heat is supplied by the element till the piston reaches the spring is 1000J

C) The ratio of work done by gas in moving the piston from initial position to A, and then from A to B is $\frac{10}{11}$

D) The ratio of work done by gas in moving the piston from initial position to A, and then from A to B is $\frac{11}{10}$

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31. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ .Its inner radius is R_1 and outer radius is R_2 . If a voltage V_0 is applied between its two ends, a current I flows in it. In addition , it is observed that transverse voltage ΔV develops between its inner and outer surface due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current).Then (figure is schematic and not drawn to scale):



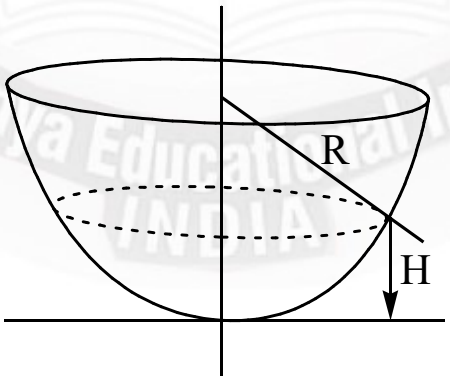
A) $I = \frac{V_0 t}{\pi\rho} \ln\left(\frac{R_2}{R_1}\right)$

B) the outer surface is at a higher voltage than the inner surface

C) the outer surface is at a lower voltage than the inner surface

D) $\Delta V \alpha I^2$

32. A small marble of mass m moves in a circular orbit very close to the bottom of a circular bowl. Assume that the marble can only move in orbits whose angular momentum about axis of the bowl is $L = \frac{nh}{2\pi}$ where h is plank constant.



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A) The allowed radius of the circular orbit is proportional to $n^{3/2}$.

B) The allowed speed is proportional to $n^{1/2}$.

C) Taking gravitational potential energy at bottom to be zero, the allowed energy is given by $\frac{nh}{2\pi}\sqrt{\frac{g}{R}}$.

D) Gravitational potential energy is equal to kinetic energy in any allowed state. (taking reference at bottom).

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. A disc placed on a frictionless horizontal plank is rotating with a constant angular velocity ω about its central vertical axis. Now the plank is made to move with a constant acceleration a on straight path. If you assume initial location of the centre of disc as origin, direction of motion of the plank is negative y-axis of the coordinate system attached with the plank, which of the following equations represent trajectory of instantaneous centre of rotation of the disc relative to plank.

A) If $\vec{\omega}=\omega\hat{k}$, then $y=\frac{\omega^2x^2}{2a}$ and $x\le 0$.

B) If $\vec{\omega}=-\omega\hat{k}$, then $y=\frac{\omega^2x^2}{a}$ and $x\ge 0$.

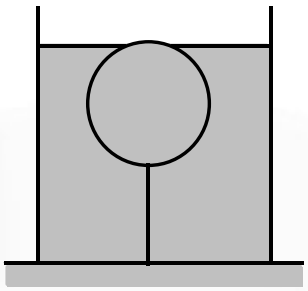
C) If $\vec{\omega}=\omega\hat{k}$, then $y=-\frac{\omega^2x^2}{2a}$ and $x\le 0$.

D) If $\vec{\omega}=-\omega\hat{k}$, then $y=-\frac{\omega^2x^2}{2a}$ and $x\ge 0$.

34. A piece of ice tied with the help of a thread to bottom of a cylindrical container having some amount of water. Length of the thread is so adjusted that the ice is not fully submerged. Tension force in the thread is 1.0 N. How much and in what way will the water level in the container shift, when all the ice melts? Area of the bottom of the vessel is $400\,cm^2$, density of water is $1.0\,g/cm^3$ and density of ice is $0.9\,g/cm^3$.

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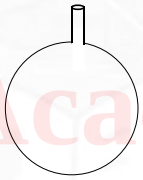

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A) $2.5\times10^{-3}\,m$ upwards **B)** $2.5\times10^{-3}\,m$ downwards

C) $2.8\times10^{-3}\,m$ upwards **D)** $2.8\times10^{-3}\,m$ downwards

35. A heat-insulated, rigid flask is evacuated and its opening is sealed with a cork. The flask is held at rest in a large chamber filled with an ideal mono-atomic gas at temperature $27^{o}C$. The cork is suddenly removed and air quickly fills the flask. If volume of the flask is negligible as compared to the volume of the chamber, what is the temperature of the air inside the flask immediately after a thermodynamic equilibrium is attained in the flask?



A) $-93^{o}C$ **B)** $27^{o}C$ **C)** $227^{o}C$ **D)** $627^{o}C$

36. A conducting shell of radius R has charge Q. Electrostatic force between two parts of the shell which are on either sides of a plane that is at a distance $r(r<R)$ from the centre of the shell is

A) $\dfrac{Q^2}{32\pi\varepsilon_0 R^2}\left(1+\dfrac{r}{R}\right)$ **B)** $\dfrac{Q^2}{32\pi\varepsilon_0 R^2}\left(1-\dfrac{r}{R}\right)$

C) $\dfrac{Q^2}{32\pi\varepsilon_0 R^2}\left(1+\dfrac{r^2}{R^2}\right)$ **D)** $\dfrac{Q^2}{32\pi\varepsilon_0 R^2}\left(1-\dfrac{r^2}{R^2}\right)$

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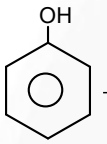
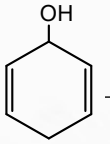
CHEMISTRY **Max Marks: 60**

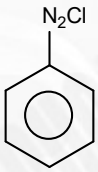
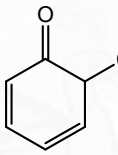
SECTION-I
(INTEGER ANSWER TYPE)

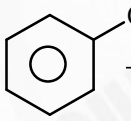
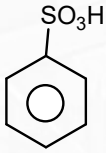
- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

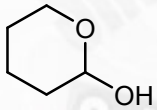
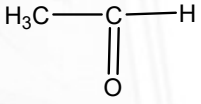
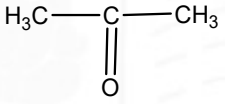
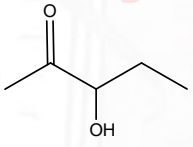
37. How many of the following reactions can give Benzene as final product

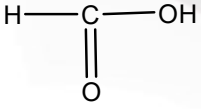
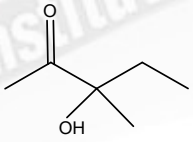
i)  ii)  iii) $CHCl_3 \xrightarrow[ii)\,\text{Red hot Fe tube}]{i)\,6Ag/\Delta}$

iv)  $\xrightarrow{Pd-C/\Delta}$ v)  $\xrightarrow{H_3PO_2}$ vi)  $\xrightarrow[\Delta]{NaOH+CaO}$

vii)  $\xrightarrow[\Delta]{NaOH+CaO}$ viii)  $\xrightarrow[ii)\,H_3O^+ \\ iii)\,Zn/\Delta]{i)\,NaOH\ (fusion)}$

38.

i)  ii)  iii)  iv)  v) $CH_3-C\equiv C-CH_3$

vi)  vii)  viii) CH_3-CH_2-OH ix) 

Number of compounds that gives Tollen's test = x

Number of compounds that gives Fehling's test = y

Then, find the value of x + y = _____

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39. A photon is emitted in β – line of Balmer series of He^+ ion. This photon is absorbed by an electron present in ground state of hydrogen atom. Calculate the value of z.

Where $z = \dfrac{\lambda}{0.529\pi}$

λ = De Broglie wave length associated with electron in final orbit of hydrogen atom (in $\overset{0}{A}$)

40. A solution contains 0.01 M concentration each of A^{2+}, B^{2+}, C^{2+} and D^{2+} ions, they are to be precipitated as their sulphide by passing H_2S gas (0.1M) into the solution.

$K_{sp(AS)} = 10^{-18}, K_{sp(DS)} = 10^{-22}, K_{sp(BS)} = 10^{-20}$

$K_{sp(CS)} = 10^{-16}, K_{a(H_2S)} = 10^{-21}$

The minimum pH required to prevent the precipitation of AS, BS, CS and DS are x, y, z and p respectively then $x + y + z - p$ is

41. Solubility constant of a sparingly soluble salt MCl_2 is 4×10^{-12} at $25^0 C$. Also, at $25^0 C$, solubility of MCl_2 in an aqueous solution of $CaCl_2$ is 4×10^8 times less compared to its solubility in pure water. Hence, concentration (molarity) of $CaCl_2$ solution is

42. Hydrolysis and condensation of three molecules of $(CH_3)_2 SiCl_2$, forms a cyclic silicone. The total number of sp^3 hybrid atoms in this silicone is _____.

(Consider the Group-14 elements only)

43. Consider the following observations:

I) a = The maximum number of identical 'N – O' bond lengths in the molecular structure of N_2O_5 in gas phase.
II) x = The maximum number of 'Cr – O' bonds with identical bond lengths in $Cr_2O_7^{2-}$.

Then the value of (x – a) is_____.

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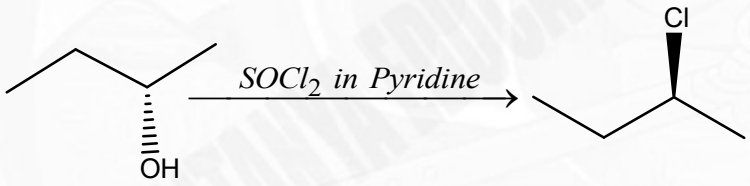
44. If $CuSO_4$ solution is added to excess KI solution, brown coloured compound 'x' is obtained as one of the product. If excess chlorine water is added to 'x', a colourless compound 'y' is obtained as one of the product. Both x and y are iodine containing products. Oxidation state of iodine in 'y' is 'a' and the average oxidation state of iodine in 'x' is 'b', then the value of $\left[(a\times b)\times(-3)\right]$ is____.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

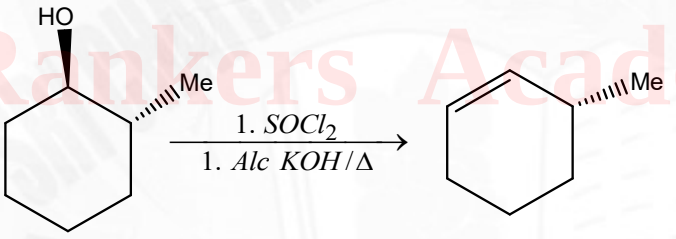
•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

45. Identify reactions correctly match with their products?

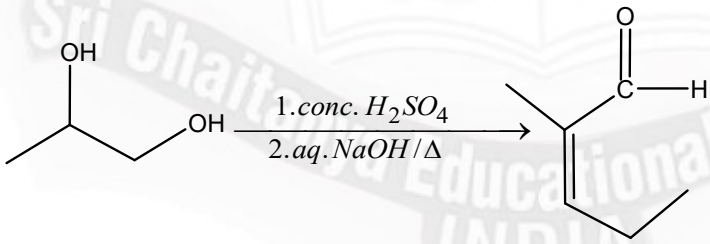
A)



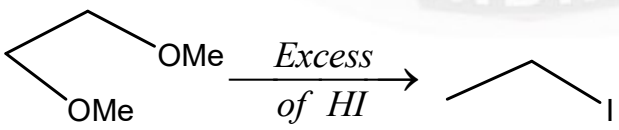
B)



C)

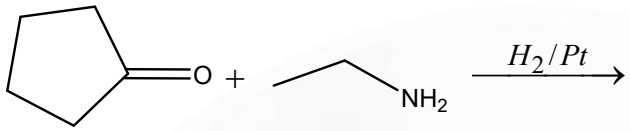


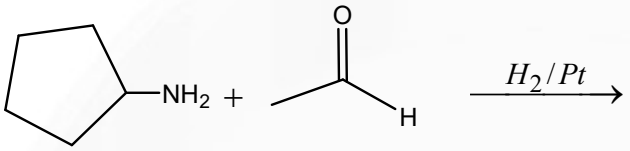
D)

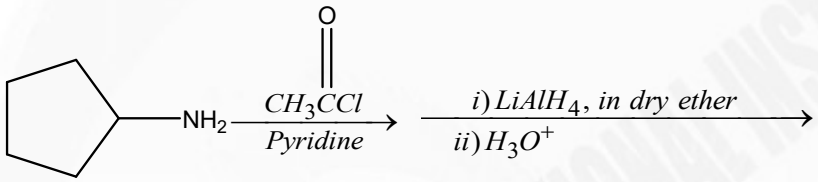


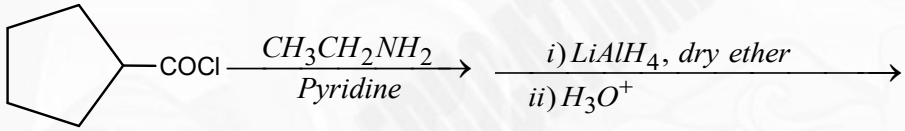
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46. Among the following, which reaction will give N-ethylcylopentylamine as major product

A) 

B) 

C) 

D) 

47. Which of the following statements are correct:-

A) Packing fraction of diamond is 0.34 approximately

B) For ideal NaCl structure $4r_{Cl^-} = a\sqrt{2}$

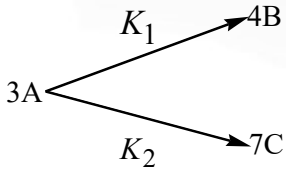
C) For ideal NaCl structure $2r_{Cl^-} + 2r_{Na^+} = a$

D) For ideal NaCl structure $\frac{r_{Na^+}}{r_{Cl^-}} = 0.414$

48. Select the correct statement(s) regarding the parallel reactions, in which both the reactions are of first order.

Given at t = 0, number of moles of A is 10; also given that $K_1 / K_2 = 3/2$.

Find the number of moles of B & C formed at $t = t_{1/2}$ *of* A.



A) number of moles of B is 2.0 **B)** number of moles of B is 4

C) number of moles of C is 10/3 **D)** number of moles of C is 14/3

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49. Which of the following is/are incorrect? (Δ_0 = Crystal field splitting energy and CFSE = Crystal field stabilisation energy)

A) $\left[Co(en)_3\right]^{3+} > \left[Co\left(NH_3\right)_6\right]^{3+} : \Delta_0$

B) $\left[Fe(ox)_3\right]^{3-} > \left[Fe(CN)_6\right]^{3-} : \Delta_0$

C) Paramagnetic $\left[NiCl_2\left(PPh_3\right)_2\right] : CFSE = -0.8\Delta_0$

D) $\left[Co\left(NH_3\right)_6\right]^{3+} > \left[Co\left(NH_3\right)_6\right]^{2+} : \Delta_0$

50. Which of the following statement(s) is/are correct?

A) Total number of $\pi -$ bonds in a molecule of $Fe_2(CO)_9$ is 18.

B) $\left[V(CO)_6\right]^- > \left[Mn(CO)_6\right]^+ > \left[Cr(CO)_6\right]$ is the correct order of CO bond strength.

C) Wilkinson catalyst is a square planar complex.

D) cis – platin is a square planar complex.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

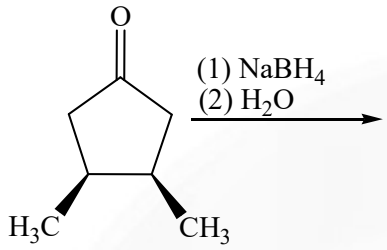
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

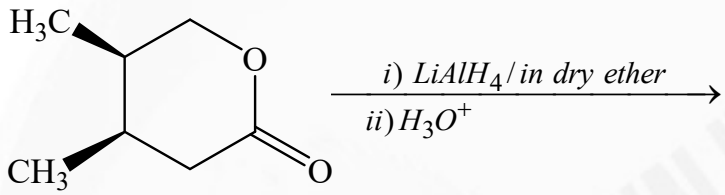
51. Find the total no of optically active aldohexoses that give optically inactive aldaric acids on nitric acid oxidation?

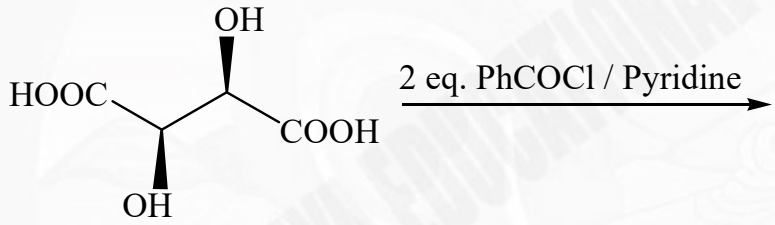
A) 2 **B)** 3 **C)** 4 **D)** 5

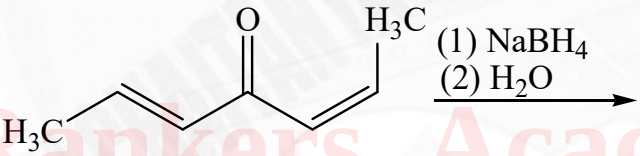
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52. Identify the reaction in which diastereomers are produced as major product?

A) 

B) 

C) 

D) 

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53. A container at 300K contains a mixture of 2 mole He, 1 mole CH_4 and 1 mole SO_2, at a total pressure of 6 atm. The container has a tiny hole through which the gases effuse. Then calculate the mole fraction of He gas in the stream of effused gas mixture?

A) 0.27 **B)** 0.73 **C)** 0.6 **D)** 0.4

54. Which of the following statements is correct?

A) H_2S turns lead acetate paper black due to the formation of lead sulphite.

B) PbS, CuS and As_2S_3 are all black coloured compounds.

C) Chromyl chloride is basic in nature.

D) Reduction of NO_3^- with Devardas alloy and NaOH gives NH_3 as one of the product.

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30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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